Plant code SG-57100-20-01

Solar ESS Co. Ltd.
53 Moo 2 Tambon Nanglae, Amphoe Muang, Chiang Rai 57100
Tel 063 151 6694 · 089 632 5829
www.solaress.org · www.โซลาร์เอสส์.com

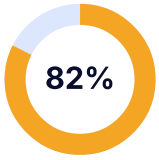
PLANT PERFORMANCE REPORT

Ban Doo Chiangrai

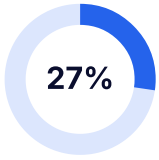
20 kWp on-grid rooftop system · no battery · zero-export setting

PERIOD	DATA SOURCE	ISSUED
23 Sep – 22 Oct 2026 (30 days)	Plant monitoring platform, daily plant report	31 Oct 2026

Solar yield	Average per day	Site consumption	Bought from grid	Sent to grid	Value of solar
1,109.9 kWh	37.0 kWh/day	3,346 kWh	2,294.7 kWh	196.4 kWh	5,217 THB @ 4.7/kWh



of solar was used on site

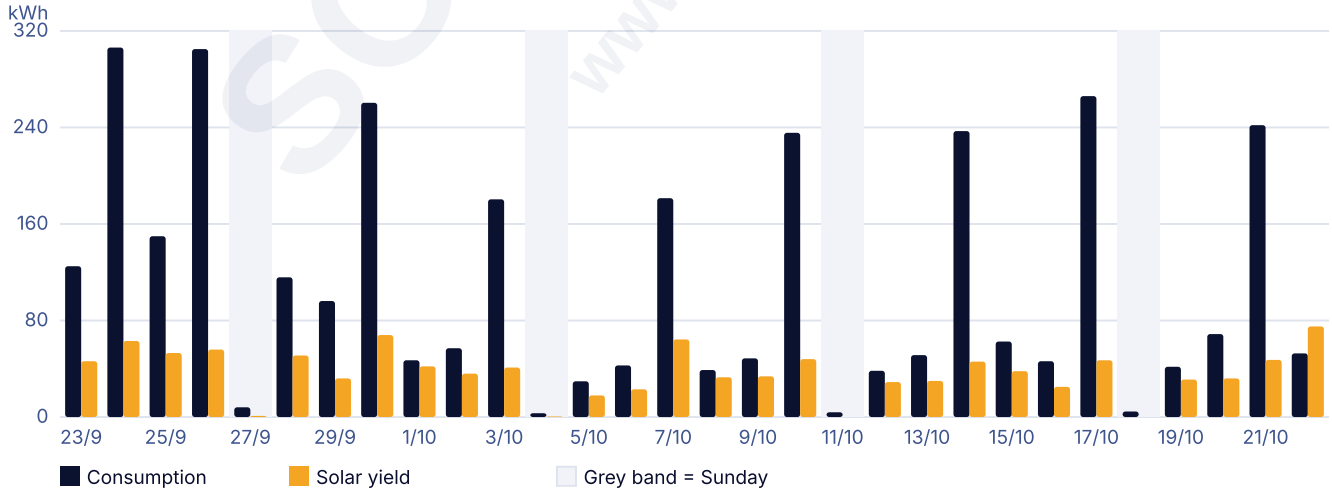


of the site's load was met by solar

Summary

- The system produced **1,109.9 kWh** in 30 days, worth about **5,217 THB** at 4.7 THB/kWh.
- **82%** of the solar energy was used directly on site; **27%** of the site's total consumption came from solar.
- Average output was **1.85 kWh per kWp per day**. This is low for a 20 kWp array, mainly because the zero-export setting holds the inverter back whenever the site load is small, and because this was the late rainy season.

Daily solar yield and consumption

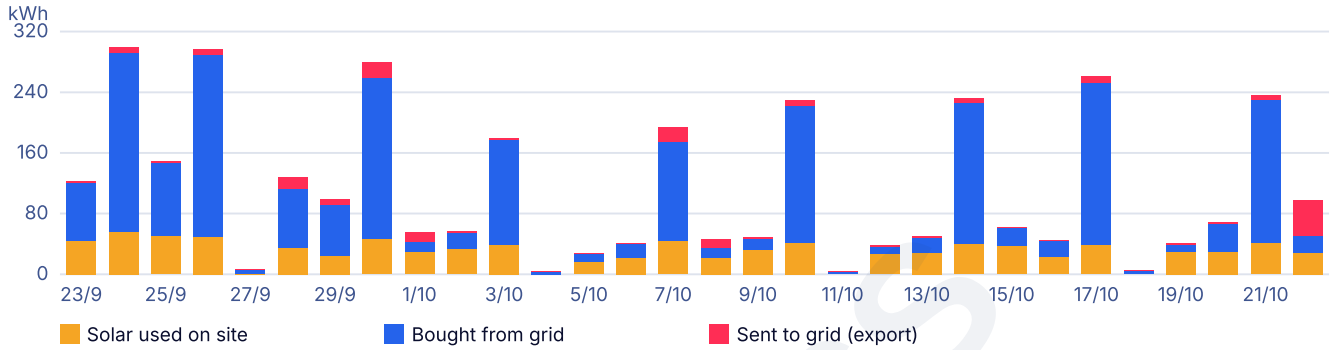


Solar yield stays fairly steady at 30–75 kWh on working days, while consumption swings between about 40 kWh and 300 kWh. On Sundays the site is almost idle and solar output falls to near zero.

SOLAR ESS www.solaress.org

Solar ESS Co. Ltd.
53 Moo 2 Tambon Nanglae, Amphoe Muang, Chiang Rai 57100
Tel 063 151 6694 · 089 632 5829
www.solaress.org · www.โซลาร์เอสส์.com

Where the site's electricity came from, day by day



Each bar is one day's total consumption, split into solar used on site, energy bought from the grid, and the small amount sent to the grid.

Three kinds of day

Day type	Days	Avg. consumption	Avg. solar yield	Solar share
Heavy-load days (machines running)	9	246.0 kWh	53.4 kWh	18%
Normal working days	17	65.4 kWh	36.9 kWh	46%
Sundays (site idle)	4	5.0 kWh	0.3 kWh	1%

Findings

1. **Export is not yet at zero.** Total export was 196.4 kWh (about 18% of yield). Most days are only 1–8 kWh, but on **22 Oct** 46.6 kWh was exported, the highest of the period. The export limit setting should be re-checked so that it is fully applied.
2. **Heavy-load days come in a cycle.** Every 3–4 days consumption rises to about 246 kWh. On those days solar covers only about 18% of the load and most energy is bought from the grid, which suggests the machine load runs largely outside peak sun hours or exceeds the array's output.
3. **Solar is held back when the load is small.** On Sundays the site uses only about 5.0 kWh and the inverter produces almost nothing (0.3 kWh/day), because zero-export stops it from producing more than the site can use. On normal days the same effect keeps yield near 37 kWh.
4. **Phase balance still matters.** The site has a history of strong phase imbalance. A balanced load across the three phases lets the inverter supply more of the daytime load under the zero-export setting.

Recommendations

1. Re-check the export limit setting and the meter/CT readings, and compare export again over the next 2 weeks.
2. Where possible, schedule the heavy machine runs between 09:00 and 15:00 to use more solar and buy less from the grid.
3. Continue to rebalance single-phase loads across the three phases.
4. If more daytime use is not possible, consider a battery to store the solar that is currently being held back, especially on Sundays and light days.

Solar ESS Co. Ltd.
53 Moo 2 Tambon Nanglae, Amphoe Muang, Chiang Rai 57100
Tel 063 151 6694 · 089 632 5829
www.solaress.org · www.โซลาร์เอสส์.com

Daily data

Date	Weather	Yield kWh	Full-load hours	Consumption kWh	Import kWh	Export kWh	Solar used kWh	Solar share	Value THB
Wed 23 Sep	Few clouds	46.2	2.31	125.0	77.1	2.23	44.0	35%	217
Thu 24 Sep	Clear	63.0	3.15	306.2	236.4	7.37	55.6	18%	296
Fri 25 Sep	Clear	53.0	2.65	149.8	95.7	2.03	51.0	34%	249
Sat 26 Sep	Light rain	56.0	2.80	304.9	240.2	6.96	49.0	16%	263
Sun 27 Sep	Light rain	1.0	0.05	8.1	6.2	0.72	0.3	3%	5
Mon 28 Sep	Light rain	51.0	2.55	115.8	76.6	15.33	35.7	31%	240
Tue 29 Sep	Light rain	32.0	1.60	96.1	67.3	7.72	24.3	25%	150
Wed 30 Sep	Light rain	68.0	3.40	260.4	211.1	20.75	47.2	18%	320
Thu 01 Oct	Light rain	42.0	2.10	47.0	13.8	12.14	29.9	64%	197
Fri 02 Oct	Light rain	36.0	1.80	57.0	20.4	1.68	34.3	60%	169
Sat 03 Oct	Few clouds	41.0	2.05	180.4	138.4	2.17	38.8	22%	193
Sun 04 Oct	Light rain	0.4	0.02	3.2	3.4	0.56	0.0	0%	2
Mon 05 Oct	Light rain	17.9	0.90	29.6	10.6	0.92	17.0	57%	84
Tue 06 Oct	Light rain	23.0	1.15	42.8	18.9	1.35	21.6	51%	108
Wed 07 Oct	Light rain	64.3	3.22	181.4	130.4	19.60	44.7	25%	302
Thu 08 Oct	Few clouds	32.9	1.65	38.9	13.6	11.38	21.5	55%	155
Fri 09 Oct	Cloudy	33.8	1.69	48.7	14.6	1.83	32.0	66%	159
Sat 10 Oct	Clear	48.0	2.40	235.5	181.9	6.78	41.2	18%	226
Sun 11 Oct	Few clouds	0.0	0.00	4.0	3.9	0.57	0.0	0%	0
Mon 12 Oct	Light rain	29.0	1.45	38.4	8.6	1.43	27.6	72%	136
Tue 13 Oct	Cloudy	30.0	1.50	51.4	20.4	1.50	28.5	55%	141
Wed 14 Oct	Few clouds	46.0	2.30	237.0	186.4	5.45	40.5	17%	216
Thu 15 Oct	Cloudy	38.0	1.90	62.6	23.6	1.23	36.8	59%	179
Fri 16 Oct	Few clouds	25.0	1.25	46.2	20.1	1.27	23.7	51%	118
Sat 17 Oct	Light rain	47.0	2.35	266.0	213.2	7.94	39.1	15%	221
Sun 18 Oct	Light rain	0.0	0.00	4.6	4.8	0.55	0.0	0%	0
Mon 19 Oct	Few clouds	31.0	1.55	41.6	10.2	1.63	29.4	71%	146
Tue 20 Oct	Clear	32.0	1.60	68.8	35.8	1.50	30.5	44%	150
Wed 21 Oct	Clear	47.3	2.37	241.9	189.1	5.24	42.1	17%	222
Thu 22 Oct	Clear	75.1	3.76	52.7	22.1	46.61	28.5	54%	353
Total		**1,109.9**	**1.85***	**3,346.0**	**2,294.7**	**196.44**	**913.5**	**27%**	**5,217**

* average per day · heavy-load day · Sunday

Notes: Solar used on site = yield − export. Solar share = solar used on site ÷ consumption. Value uses the platform tariff of 4.7 THB/kWh. Weather as recorded by the monitoring platform. Small export on idle days (above yield) is within meter tolerance.